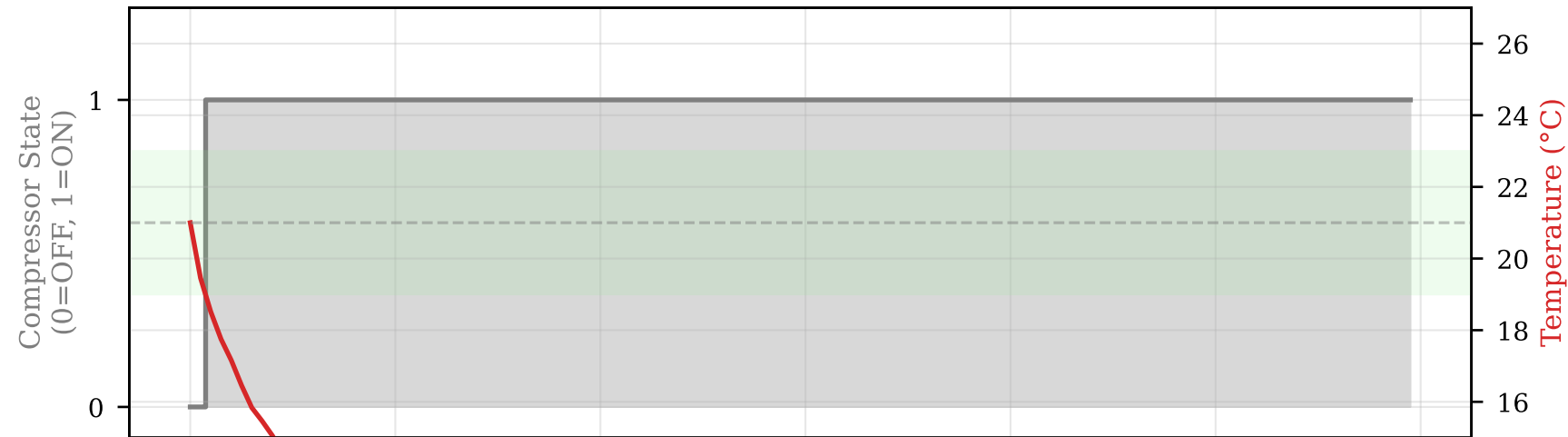
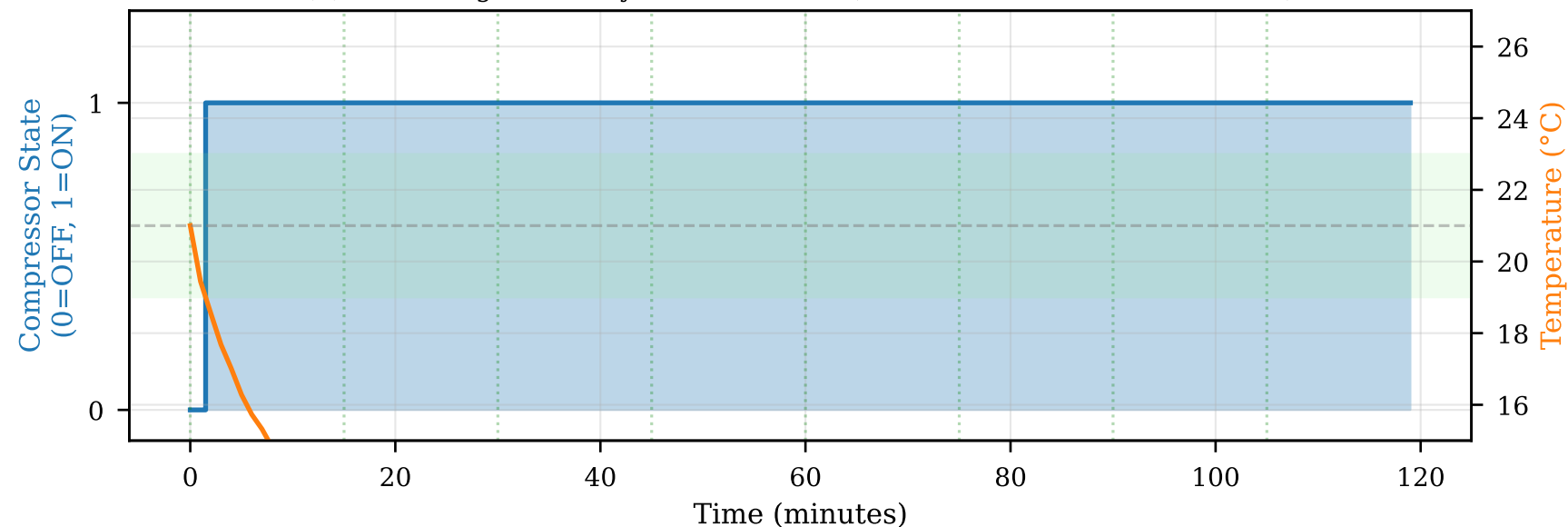


System Heartbeat: Micro-Dynamics Comparison (Prevention of Short-Cycling)

(a) Baseline Thermostat — 1 cycles in 120 min (Short-cycling highlighted in red)



(b) PI-DRL Agent — 1 cycles in 120 min (Green lines: 15-min boundaries)



- Compressor ON
- Baseline Temp
- PI-DRL Temp
- Comfort Zone
- Short-Cycling Event